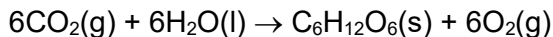


11 Some $\Delta H_f^\ominus$ values are given below.

compound	$\Delta H_f^\ominus / \text{kJ mol}^{-1}$
$\text{H}_2\text{O}(\text{l})$	-286
$\text{CO}_2(\text{g})$	-394
$\text{C}_6\text{H}_{12}\text{O}_6(\text{s})$	-1273

The overall reaction in photosynthesis can be represented by the following equation.



Which of the following statements are correct?

- 1 The enthalpy change of the reaction is $+2807 \text{ kJ mol}^{-1}$
- 2 In the formation of products, the system becomes less disordered.
- 3 The reaction is not spontaneous at all temperatures.

A 1, 2 and 3 **B** 1 and 2 only **C** 2 and 3 only **D** 1 only